

BABU BANARASI DAS
UNIVERSITY

2026-2027



**SCHOOL OF COMPUTER
APPLICATIONS**

No SQL Project On

**Campus Lost And Found System in
MongoDB**

Submitted to:

Submitted by:

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Department: BCA (DS & AI)

Date: 25th September

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1. CREATE DATABASE

MongoDB Query:

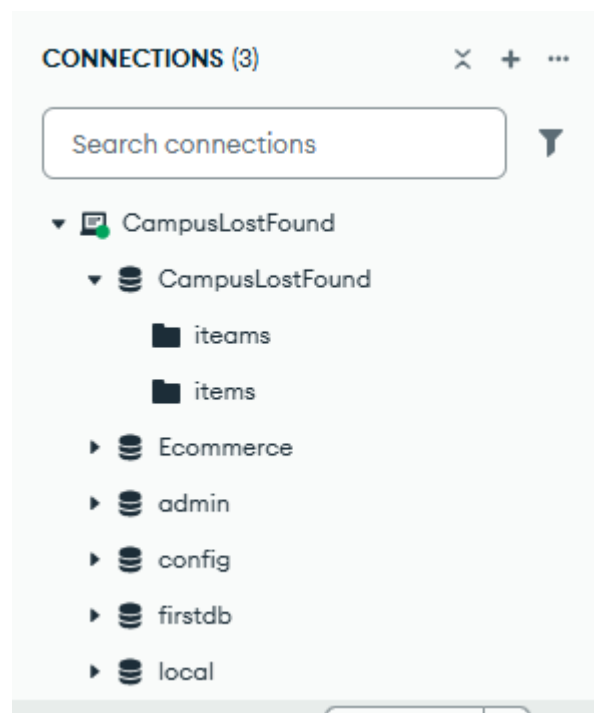
Use campusLostFound

```
>_MONGOSH  
  
> use("CampusLostFound")  
< switched to db CampusLostFound
```

2.CREATE COLLECTION

MongoDB Query:

db.create collection("CampusLostFound")



3. INSERT 50 ITEMS

MongoDB Query:

```
db.items.insertMany([ { itemId:101, itemName:"Black Wallet", category:"Personal",
location:"Library", dateReported:ISODate("2026-09-01T10:00:00Z"), reportedBy:{name:"Aarav
Sharma",department:"BCA"}, status:"Found", estimatedValue:450, tags:["black","leather","wallet"] },

{ itemId:102, itemName:"Blue Water Bottle", category:"Accessories", location:"Block A",
dateReported:ISODate("2026-09-01T10:00:00Z"), reportedBy:{name:"Priya
Singh",department:"BCA"}, status:"Claimed", estimatedValue:650, tags:["blue","bottle","steel"] },

{ itemId:103, itemName:"Scientific Calculator", category:"Study Material", location:"Lab 2",
dateReported:ISODate("2026-09-02T10:00:00Z"), reportedBy:{name:"Rohan
Verma",department:"BCA"}, status:"Found", estimatedValue:900, tags:["calculator","casio","black"]
},

{ itemId:104, itemName:"USB Drive 32GB", category:"Electronics", location:"Computer Lab",
dateReported:ISODate("2026-09-02T10:00:00Z"), reportedBy:{name:"Ananya
Gupta",department:"BCA"}, status:"Returned", estimatedValue:700, tags:["usb","pendrive","32gb"]
},

{ itemId:105, itemName:"Black Umbrella", category:"Personal", location:"Main Gate",
dateReported:ISODate("2026-09-03T10:00:00Z"), reportedBy:{name:"Vivek
Kumar",department:"BCA"}, status:"Found", estimatedValue:350, tags:["umbrella","black"] },

{ itemId:106, itemName:"Silver Bracelet", category:"Accessories", location:"Cafeteria",
dateReported:ISODate("2026-09-03T10:00:00Z"), reportedBy:{name:"Meera
Singh",department:"BBA"}, status:"Found", estimatedValue:1200, tags:["bracelet","silver"] },
```

```
> _MONGOSH
```

```
> use("CampusLostFound")
```

```
< switched to db CampusLostFound
```

```
> db.items.insertMany([
```

```
  {itemId:101,itemName:"Black Wallet",category:"Personal",location:"Library",date:
```

```
  {itemId:102,itemName:"Blue Water Bottle",category:"Accessories",location:"Block
```

```
  {itemId:103,itemName:"Scientific Calculator",category:"Study Material",location:
```

```
  {itemId:104,itemName:"USB Drive 32GB",category:"Electronics",location:"Computer
```

```
  {itemId:105,itemName:"Black Umbrella",category:"Personal",location:"Main Gate",
```

```
  {itemId:106,itemName:"Silver Bracelet",category:"Accessories",location:"Cafeter
```

```
  {itemId:107,itemName:"Notebook DBMS",category:"Study Material",location:"Classr
```

```
  {itemId:108,itemName:"AirPods Case",category:"Electronics",location:"Library",c
```

```
  {itemId:109,itemName:"Grey Hoodie",category:"Clothing",location:"Sports Ground"
```

```
  {itemId:110,itemName:"ID Card Holder",category:"Documents",location:"Block B",c
```

```
  {itemId:111,itemName:"Red Backpack",category:"Bags",location:"Parking Area",dat
```

```
  {itemId:112,itemName:"Power Bank",category:"Electronics",location:"Cafeteria",c
```

```
  {itemId:113,itemName:"Blue Pen Set",category:"Study Material",location:"Classro
```

```
  {itemId:114,itemName:"Black Spectacles",category:"Accessories",location:"Semin
```

```
  {itemId:115,itemName:"Lunch Box",category:"Personal",location:"Cafeteria",date
```

```
  {itemId:116,itemName:"Keyboard",category:"Electronics",location:"Computer Lab"
```

```
  {itemId:117,itemName:"Java Notes",category:"Study Material",location:"Library"
```

```
  {itemId:118,itemName:"Green Cap",category:"Clothing",location:"Sports Ground",
```

```
  {itemId:119,itemName:"Maths Textbook",category:"Books",location:"Library",date
```

```
  {itemId:120,itemName:"Charging Cable",category:"Electronics",location:"Hostel B
```

```
  {itemId:121,itemName:"Black Shoes",category:"Clothing",location:"Sports Ground"
```

4. DISPLAY ALL CAMPUSLOSTFOUND

MongoDB Query:

```
db.item.find()
```

```

<
> db.items.find()
< {
  _id: ObjectId('6ab15ea5d260b87d9603e881'),
  itemId: 101,
  itemName: 'Black Wallet',
  category: 'Personal',
  location: 'Library',
  dateReported: ISODate('2026-09-01T00:00:00.000Z'),
  reportedBy: {
    name: 'Aarav Sharma',
    department: 'BCA'
  },
  status: 'Found',
  estimatedValue: 450,
  tags: [
    'black',
    'leather',
    'wallet'
  ]
}
{
  _id: ObjectId('6ab15ea5d260b87d9603e882'),
  itemId: 102,
  itemName: 'Blue Water Bottle',

```

5. COUNTDOCUMENT

MongoDB Query: `db.items.countDocuments()`

```

> db.items.countDocuments()
< 50

```

6. HIGHEST VALUE ITEMS

MongoDB Query:

`db.items.find().sort({estimatedValue: -1}).limit(1)`

```
> db.items.find().sort({estimatedValue: -1}).limit(1)
< {
  _id: ObjectId('6ab15ea5d260b87d9603e89e'),
  itemId: 130,
  itemName: 'Bluetooth Speaker',
  category: 'Electronics',
  location: 'Auditorium',
  dateReported: ISODate('2026-09-15T00:00:00.000Z'),
  reportedBy: {
    name: 'Kavya Singh',
    department: 'BCA'
  },
  status: 'Found',
  estimatedValue: 2500,
  tags: [
    'speaker',
    'bluetooth'
  ]
}
```

7. FIND ITEMS

MongoDB Query: db.items.find()

```

<
> db.items.find()
< {
  _id: ObjectId('6ab15ea5d260b87d9603e881'),
  itemId: 101,
  itemName: 'Black Wallet',
  category: 'Personal',
  location: 'Library',
  dateReported: ISODate('2026-09-01T00:00:00.000Z'),
  reportedBy: {
    name: 'Aarav Sharma',
    department: 'BCA'
  },
  status: 'Found',
  estimatedValue: 450,
  tags: [
    'black',
    'leather',
    'wallet'
  ]
}
{
  _id: ObjectId('6ab15ea5d260b87d9603e882'),
  itemId: 102,
  itemName: 'Blue Water Bottle',

```

8. FOUND SPECIFIC ITEM

MongoDB Query: `db.items.find({status:"Found"})`

```

> db.items.find({ status:"Found"})
< {
  _id: ObjectId('6ab15ea5d260b87d9603e881'),
  itemId: 101,
  itemName: 'Black Wallet',
  category: 'Personal',
  location: 'Library',
  dateReported: ISODate('2026-09-01T00:00:00.000Z'),
  reportedBy: {
    name: 'Aarav Sharma',
    department: 'BCA'
  },
  status: 'Found',
  estimatedValue: 450,
  tags: [
    'black',
    'leather',
    'wallet'
  ]
}
{
  _id: ObjectId('6ab15ea5d260b87d9603e883'),
  itemId: 103,
  itemName: 'Scientific Calculator',

```

9. FOUNDS ITEMS SPECIFIC

MongoDB Query: `db.items.find({location:"Library"})`

```

> _MONGOSH
> db.items.find({
  location: "Library"
})
< {
  _id: ObjectId('6ab15ea5d260b87d9603e881'),
  itemId: 101,
  itemName: 'Black Wallet',
  category: 'Personal',
  location: 'Library',
  dateReported: ISODate('2026-09-01T00:00:00.000Z'),
  reportedBy: {
    name: 'Aarav Sharma',
    department: 'BCA'
  },
  status: 'Found',
  estimatedValue: 450,
  tags: [
    'black',
    'leather',
    'wallet'
  ]
}

```

10.TOP 5 MOST VALUED ITEMS

MongoDB Query: `db.items.find({status:"Found"}).sort({estimatedValue:-1}).limit(5)`

>_MONGOSH

```
> db.items.find().sort({ estimatedValue:-1}).limit(5)
```

```
< {
  _id: ObjectId('6ab15ea5d260b87d9603e89e'),
  itemId: 130,
  itemName: 'Bluetooth Speaker',
  category: 'Electronics',
  location: 'Auditorium',
  dateReported: ISODate('2026-09-15T00:00:00.000Z'),
  reportedBy: {
    name: 'Kavya Singh',
    department: 'BCA'
  },
  status: 'Found',
  estimatedValue: 2500,
  tags: [
    'speaker',
    'bluetooth'
  ]
}
{
  _id: ObjectId('6ab15ea5d260b87d9603e895'),
  itemId: 121,
  itemName: 'Black Shoes',
  category: 'Clothing',
```

11.CREATION OPERSTION(INSERTONE)

MongoDB Query: db.items.insertOne({itemId:151,itemName:"Red pen",catrgory:"Study Material",location:"Library",status:"Found",estimedValue:50})

```

>_MONGOSH

> use("CampusLostFound")
< switched to db CampusLostFound

> db.items.insertOne({
  itemId: 151,
  itemName: "Red Pen",
  category: "Study Material",
  location: "Library",
  status: "Found",
  estimatedValue: 50
})
< {
  acknowledged: true,
  insertedId: ObjectId('6ab26216232b76361ea38c29')
}
CampusLostFound > |

```

12.INSERTMANY

MongoDB Query: db.items.insertMany([{itemId:152,itemName:"Blue Bag",category:"Bags",status:"Found"},{itemId:153,itemName:"Black Shoes",category:"Clothing",status:"Found"}])

```
> db.items.insertMany([
  {
    itemId: 152,
    itemName: "Blue Bag",
    category: "Bags",
    status: "Found"
  },
  {
    itemId: 153,
    itemName: "Black Shoes",
    category: "Clothing",
    status: "Found"
  }
])
< {
  acknowledged: true,
  insertedIds: {
    '0': ObjectId('6ab264df232b76361ea38c2a'),
    '1': ObjectId('6ab264df232b76361ea38c2b')
  }
}
```

13.FINDONE

MongoDB Query: db.items.findOne({itemId:101})

```
>_MONGOSH
}
> db.items.findOne({
  itemId: 101
})
< {
  _id: ObjectId('6ab15ea5d260b87d9603e881'),
  itemId: 101,
  itemName: 'Black Wallet',
  category: 'Personal',
  location: 'Library',
  dateReported: ISODate('2026-09-01T00:00:00.000Z'),
  reportedBy: {
    name: 'Aarav Sharma',
    department: 'BCA'
  },
  status: 'Found',
  estimatedValue: 450,
  tags: [
    'black',
    'leather',
    'wallet'
  ]
}
```

CampusLostFound >

14. FIND.SORT

MongoDB Query: `db.items.find().sort({estimatedValue:-1})`

>_MONGOSH

```
> db.items.find().sort({
  estimatedValue: -1
})
< {
  _id: ObjectId('6ab15ea5d260b87d9603e89e'),
  itemId: 130,
  itemName: 'Bluetooth Speaker',
  category: 'Electronics',
  location: 'Auditorium',
  dateReported: ISODate('2026-09-15T00:00:00.000Z'),
  reportedBy: {
    name: 'Kavya Singh',
    department: 'BCA'
  },
  status: 'Found',
  estimatedValue: 2500,
  tags: [
    'speaker',
    'bluetooth'
  ]
}
{
  _id: ObjectId('6ab15ea5d260b87d9603e895'),
  itemId: 121,
```

15. FIND.LIMIT

MongoDB Query: `db.items.find().limit(5)`

```
> db.items.find().limit(5)
< {
  _id: ObjectId('6ab15ea5d260b87d9603e881'),
  itemId: 101,
  itemName: 'Black Wallet',
  category: 'Personal',
  location: 'Library',
  dateReported: ISODate('2026-09-01T00:00:00.000Z'),
  reportedBy: {
    name: 'Aarav Sharma',
    department: 'BCA'
  },
  status: 'Found',
  estimatedValue: 450,
  tags: [
    'black',
    'leather',
    'wallet'
  ]
}
{
  _id: ObjectId('6ab15ea5d260b87d9603e882'),
  itemId: 102,
  itemName: 'Blue Water Bottle',
```

16. FIND.SKIP

MongoDB Query: `db.items.find().skip(10)`

>_MONGOSH

> db.items.find().skip(10)

```
< {
  _id: ObjectId('6ab15ea5d260b87d9603e88b'),
  itemId: 111,
  itemName: 'Red Backpack',
  category: 'Bags',
  location: 'Parking Area',
  dateReported: ISODate('2026-09-06T00:00:00.000Z'),
  reportedBy: {
    name: 'Sakshi Gupta',
    department: 'BCA'
  },
  status: 'Found',
  estimatedValue: 1800,
  tags: [
    'bag',
    'red',
    'backpack'
  ]
}
{
  _id: ObjectId('6ab15ea5d260b87d9603e88c'),
  itemId: 112,
  itemName: 'Power Bank',
```

16. PROJECTION

MongoDB Query: `db.items.find({}, {_id:0,itemName:1,status:1,})`

```
>_MONGOSH
```

```
> db.items.find(
  {},
  {
    _id: 0,
    itemName: 1,
    status: 1
  }
)
< {
  itemName: 'Black Wallet',
  status: 'Found'
}
{
  itemName: 'Blue Water Bottle',
  status: 'Claimed'
}
{
  itemName: 'Scientific Calculator',
  status: 'Found'
}
{
  itemName: 'USB Drive 32GB',
  status: 'Returned'
}
```

17.UPDATE OPERATION(Update one())

MongoDB Query: `db.items.updateOne( { itemId: 101 }, { $set: { status: "Returned" } })`

```
Type "it" for more
> db.items.updateOne(
  { itemId: 101 },
  { $set: { status: "Returned" } }
)
< {
  acknowledged: true,
  insertedId: null,
  matchedCount: 1,
  modifiedCount: 1,
  upsertedCount: 0
}
CampusLostFound>
```

18. UPDATE MANY

MongoDB Query: `db.items.updateMany({ category: "Electronics" }, { $set:{priority:"High"}})`

```
> db.items.updateMany(
  { category: "Electronics" },
  { $set: { priority: "High" } }
)
< {
  acknowledged: true,
  insertedId: null,
  matchedCount: 10,
  modifiedCount: 10,
  upsertedCount: 0
}
CampusLostFound>
```

19. REPLACEONE()

MongoDB Query: `db.items.replaceOne({ itemId: 151 },{{ itemId: 151 ,itemName:"Red Pen",category:"Study Material",status:"Returned:}}`

```

> db.items.replaceOne(
  { itemId: 151 },
  {
    itemId: 151,
    itemName: "Red Pen",
    category: "Study Material",
    status: "Returned"
  }
)
< {
  acknowledged: true,
  insertedId: null,
  matchedCount: 1,
  modifiedCount: 1,
  upsertedCount: 0
}
CampusLostFound>

```

20. DELETE OPERATION(Deleteone)

MongoDB Query: db.items.deleteOne({itemId:151})

```

> db.items.deleteOne({
  itemId: 151
})
< {
  acknowledged: true,
  deletedCount: 1
}
CampusLostFound>

```

21.DELETE MANY

MongoDB Query: db.items.deleteMany({status:"Clamined"})

```

> db.items.deleteMany({
  status: "Clamined"
})
< {
  acknowledged: true,
  deletedCount: 9
}
CampusLostFound>

```

22.DISTINCT

MongoDB Query: `db.items.distinct("category")`

```
> db.items.distinct("category")
< [
  'Accessories',
  'Bags',
  'Books',
  'Clothing',
  'Documents',
  'Electronics',
  'Personal',
  'Study Material'
]
CampusLostFound > |
```

23.MAX

MongoDB Query:

`db.items.aggregate([{$group:{_id:null,maximumValue:{$max:"$estimatedValue"}}})`

```
> db.items.aggregate([
  {
    $group: {
      _id: null,
      maximumValue: {
        $max: "$estimatedValue"
      }
    }
  }
])
< {
  _id: null,
  maximumValue: 2500
}
```

24. COUNT

MongoDB Query: `db.items.aggregate([{$count:"totalItems"}])`

```
> db.items.aggregate([
  {
    $count: "totalItems"
  }
])
< {
  totalItems: 43
}
```

25. SKIP

MoongoDB Query: db.items.aggregate([{\$skip:10}])

```
>_MONGOSH
> db.items.aggregate([
  {
    $project: {
      _id: 0,
      itemName: 1,
      status: 1
    }
  }
])
< {
  itemName: 'Black Wallet',
  status: 'Returned'
}
{
  itemName: 'Scientific Calculator',
  status: 'Found'
}
{
  itemName: 'USB Drive 32GB',
  status: 'Returned'
}
{
  itemName: 'Black Umbrella',
  status: 'Found'
}
```

